

Module 4: Energy, reproduction and populations

4.1 Energy, metabolism and exercise

This section builds on the knowledge and understanding gained in Module 2 by considering the role of ATP as an immediate source of energy for biological processes and the biochemical pathways of cellular respiration that generate ATP.

The concept of aerobic fitness is explored in the context of the effects of exercise on the respiratory and cardiovascular systems. The effects of exercise on skeletal muscles and the mechanism of muscle contraction at cellular level are also considered.

The process of cellular respiration requires an efficient supply of oxygen and the importance of respiratory pigments as oxygen carriers is illustrated by reference to haemoglobin and myoglobin. Learners also gain an insight into the unique properties of fetal haemoglobin in meeting the metabolic needs of the fetus during development.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

4.1.1 Cellular respiration

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) an outline of glycolysis as an enzyme controlled metabolic pathway	To include the location of the enzymes (only dehydrogenase to be named), the conversion of glucose to triose phosphate, the production of pyruvate, reduced NAD and the production of ATP by substrate level phosphorylation. HSW8
(b) an outline of the link reaction	To include the location of the enzymes, decarboxylation and dehydrogenation of pyruvate (3C) to acetyl (2C) coenzyme A, and the reduction of NAD. HSW8
(c) an outline of the Krebs cycle	To include the location of the enzymes, the formation of citrate from the acetyl group of acetyl CoA and oxaloacetate, the conversion of citrate to oxaloacetate using decarboxylation and dehydrogenation reactions and the production of reduced NAD and reduced FAD and also ATP by substrate level phosphorylation (names of intermediate compounds are not required). HSW8

- (d) an outline of the process of oxidative phosphorylation
- To include the location of the process and the roles of electron carriers, oxygen and the mitochondrial cristae (inner membrane)
AND
 to include the electron transport chain, proton gradients and ATP synthase.
 HSW8
- (e) an outline of the process of anaerobic respiration in muscle cells and in yeast
- To include an appreciation that anaerobic respiration produces a much lower yield of ATP than aerobic respiration.
- (f) the relative energy values of different respiratory substrates
- To include the calculation and interpretation of respiratory quotients for carbohydrates, proteins and lipids.
- $$RQ = \frac{CO_2 \text{ produced}}{O_2 \text{ consumed}}$$
- M0.3, M2.3, M2.4
 HSW8, HSW12
- (g) (i) the use of respirometers and other methods to investigate the rate of respiration
- M0.1, M0.2, M0.3, M1.1, M1.2, M1.3, M1.6, M1.9, M3.2, M3.3, M3.5, M3.6
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 HSW2, HSW3, HSW4, HSW5, HSW6
- (ii) practical investigations into the effect of temperature, substrate concentration, anaerobic conditions and different respiratory substrates on the rate of respiration.